Estd : 1984

Transform Yourself

ENSEMBLE LEARNING

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ENSEMBLE INTRODUCTION

Ensembling is a process of combining many weak learners to produce a strong learner that performs better than a single weak learner. The idea is to learn better with a set of models, instead of making one model to learn better. It is like getting opinions from many in order to decide and to protect oneself in decision-making.

KEY POINTS ON ENSEMBLE LEARNING:

Basic Ensembling Techniques:

A simple way of ensembling which involves majority voting or taking the average or weighted average of predictions from multiple machine learning models or base learners.

Stacking:

Stacking is an ensembling method that uses predictions from multiple models as "features" to train a new model, which then provides the final prediction.

Repeated Model Ensembling:

This approach involves employing the same machine learning model multiple times on different datasets and combining the hypotheses generated from each model

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BAGGING (BOOTSTRAP AGGREGATING):

Bagging involves using multiple independent models simultaneously and averaging their predictions, which makes the overall model more robust than a single classifier.
This method addresses the problem of overfitting, which often occurs with a single classifier, by minimizing the variance of the learning model.

BOOSTING:

Boosting combines different classifiers in a sequential manner, where each classifier's prediction is refined by increasing the weights of data instances that were misclassified.
Each subsequent model receives the output of the previous one, increasing the accuracy of the final predictions.
Boosting aims to improve prediction accuracy rather than reduce variance.

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Purpose of Ensemble Learning:

Ensemble learning minimizes bias and variance and helps build accurate models that avoid overfitting (fitting the training data too closely) and underfitting (missing important data patterns).

Cross-Validation in Ensemble Learning:

Cross-validation can be applied to train on multiple datasets, averaging their predictions to ensemble various models' outputs.

Fine-Tuning in Ensemble Learning:

Fine-tuning the parameters of individual models, such as tuning the number of trees or split variables in a random forest classifier, can improve model accuracy before ensembling.

Ensembling combines various fine-tuned models to enhance overall performance, though there is a tradeoff between accuracy and model complexity.

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Advantages of Ensemble Learning

1. *Minimizes bias and variance of a weak learner*
2. *Reduces overfitting and underfitting of the model*
3. *Increases accuracy of prediction*
4. *Handles higher dimensionality data efficiently*

Disadvantages of Ensemble Learning

1. *Does not give precise values for the classification and regression mode l*
2. *Error prone and sometimes leads to overfitting*
3. *Requires careful tuning of the number and type of features used*

ENSEMBLING TECHNIQUES

- *PARALLEL ENSEMBLE MODELS*
- *INCREMENTAL ENSEMBLE MODELS*
- *SEQUENTIAL ENSEMBLE MODELS*

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PARALLEL ENSEMBLE MODELS

Parallel ensemble models provide a simple way of ensembling the predictions from multiple independent models and then applying averaging techniques such as normal average or weighted average or majority voting to estimate the final prediction

Voting

It is a simple way of ensembling multiple learners. It is the process of combining multiple machine learning models or using the same model multiple times as base learners and taking average of their predictions in the case of regression problem and considering the maximum vote of their predictions in the case of classification problem.

BOOTSTRAP RESAMPLING

It is a process of creating many bootstrap samples S, (1 ≤ i ≤ M) using a method called Random Sampling with replacement from the training dataset. Each bootstrap sample contains 63.2% distinct tuples from the original training dataset and the remaining tuples are duplicate tuples drawn from the original dataset. If the training dataset T contains N tuples, M samples can be created, where each bootstrap sample S, consists of N tuples by randomly choosing N tuples from the original training dataset with replacement

BAGGING

Bagging is otherwise called as Bootstrap Aggregation which combines both bootstrapping and aggregation (Breiman, 1996). It uses Bootstrap Resampling to create M bootstrap samples from the original training dataset T. It then trains M independent weak classifiers with each classifier containing a bootstrap sample S_{i}(1 <= i <= M) Then, it combines the predictions made by M weak classifiers either by majority voting variable is a categorical variable or by averaging if the target variable is a continuous variable. By combining or ensembling many classifiers predictions, the model reduces the error or decreases the variance, resulting in the prediction of a single weak classifier.

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BAGGING ALGORITHM

Input: Training dataset T with N data instances.

Step 1: Create M bootstrap samples $S_i (1 \le i \le M)$ of N data instances each from the training dataset T using Random sampling with Replacement.

Step 2: Train M weak classifiers H_i $(1 \le i \le M)$ using the samples $S_i (1 \le i \le M)$.

Step 3: Combine predictions from all classifiers taking average $H_{ave} = \frac{1}{N}\sum_{i=1}^{M} H_i$.

RANDOM FOREST

Random Forest is an ensemble model that uses Bagging with decision trees as its base learners. It generates M bootstrap samples from an initial training dataset of N tuples by random sampling with replacement, which are then used to train M individual decision trees. Unlike traditional Bagging, Random Forest selects a random subset of features for each tree, adding variability. At each node within these trees, a random split feature is chosen, enhancing differentiation among trees within the ensemble.

This model uses majority voting or averaging across the trees for final predictions, leading to improved accuracy and reduced variance. Random Forest's diversity, created by using varied samples, features, and split criteria, makes it robust and efficient, as shown in Figure 12.1.

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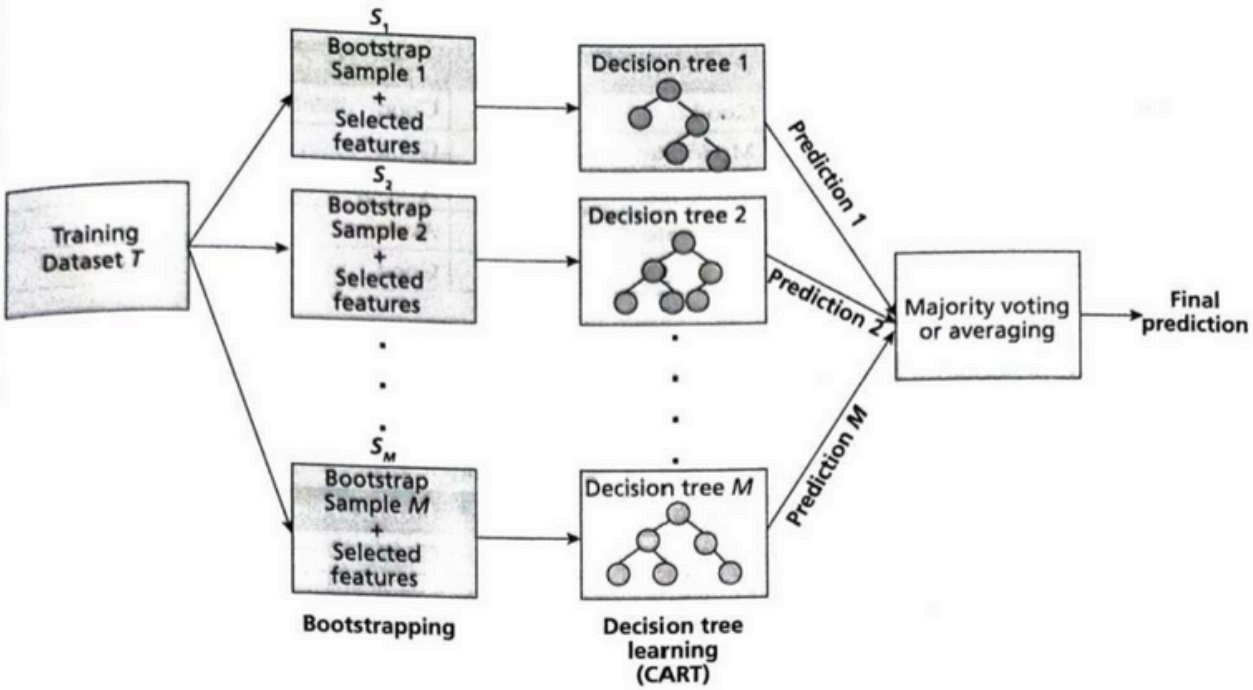

Figure 12.1: Random Forest Model

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RANDOM FOREST ALGORITHM

Input: Training dataset T with N data instances.

Step 1: Create M bootstrap samples S_i $(1 \le i \le M)$ of N data instances each from the training dataset T using Random sampling with Replacement.

Step 2: Choose number of Decision trees to be constructed DT_M.

Step 3: Choose the number of features to be used in each tree. Normally, $\sqrt{D}$ randomly chosen features are taken.

Step 4: Choose the depth of the decision tree. All trees are to be same depth.

Step 5: Repeat M times. For each tree:

(a) Choose a bootstrap sample S_i of N data instances.

(b) Compute the best split feature to be placed at each node.

(c) Construct decision tree using CART. Do not prune the tree.

Step 6: Combine predictions from all classifiers taking Average $H_{ave} = \frac{1}{N}\sum_{i=1}^{M} H_i$ for classification problem or Majority voting in the case of regression problem.

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Example 12.1: Consider a training dataset of 5 data instances shown in Table 12.1 which describes the performance of students and their skills obtained during study. Based on their performance and the skills obtained, they would get a Job Offer at the end of the course. Train using a Random Forest model and test this ensemble model.

Table 12.1: Training Dataset

S.No.	CGPA	Interactiveness	Communication Skills	Practical Knowledge	Job Offer
1.	≥9	Yes	Good	Good	Yes
2.	<9	No	Moderate	Good	Yes
3.	≥9	No	Moderate	Average	No
4.	<9	No	Moderate	Average	No
5.	≥9	Yes	Moderate	Good	Yes

Solution:

Step 1: Create 3 bootstrap samples of 5 data instances each from the training dataset T using random sampling with replacement as shown in Tables 12.2, 12.3 and 12.4, respectively.

Table 12.2: Sample S_1

S.No.	CGPA	Interactiveness	Communication Skills	Practical Knowledge	Job Offer
1.	≥9	Yes	Good	Good	Yes
2.	<9	No	Moderate	Good	Yes
3.	≥9	No	Moderate	Average	No
3.	≥9	No	Moderate	Average	No
5.	≥9	Yes	Moderate	Good	Yes

Table 12.3: Sample S_2

S.No.	CGPA	Interactiveness	Communication Skills	Practical Knowledge	Job Offer
2.	<9	No	Moderate	Good	Yes
3.	≥9	No	Moderate	Average	No
3.	≥9	No	Moderate	Average	No
5.	≥9	Yes	Moderate	Good	Yes
5.	≥9	Yes	Moderate	Good	Yes

Table 12.4: Sample S_3

S.No.	CGPA	Interactiveness	Communication Skills	Practical Knowledge	Job Offer
1.	≥9	Yes	Good	Good	Yes
1.	≥9	Yes	Good	Good	Yes
2.	<9	No	Moderate	Good	Yes
3.	≥9	No	Moderate	Average	No
3.	≥9	No	Moderate	Average	No

Step 2: Choose number of Decision trees to be constructed as 3.

Step 3: Choose the number of features to be used in each tree. Here, we will choose 2 features to construct every decision tree.

Step 4: Choose the depth of each decision tree to be 2.

Step 5: Repeat the following steps for 3 decision trees that are constructed:

Step 5 (a): Choose bootstrap sample S_1 of 5 data instances to construct Decision tree 1.

Step 5 (b): Compute the best split feature to be placed at each node. Here, we choose CGPA and Interactiveness.

Step 5 (c): Construct decision tree using CART shown in Figure 12.2.

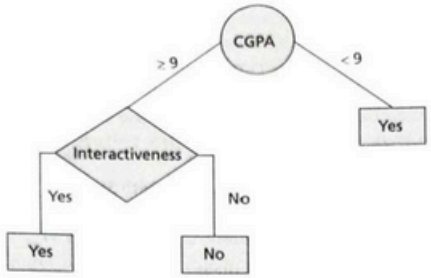

Figure 12.2: Decision Tree 1 with Sample S_1

Step 5 (a): Choose bootstrap sample S_2 of 5 data instances to construct Decision tree 2.

Step 5 (b): Compute the best split feature to be placed at each node. Here, we choose Interactiveness and Practical Knowledge.

Step 5 (c): Construct decision tree using CART shown in Figure 12.3.

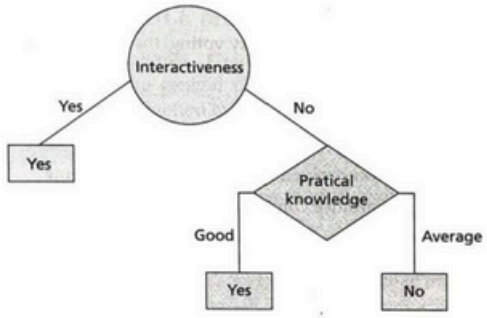

Figure 12.3: Decision Tree 2 with Sample S_2

Step 5 (a): Choose bootstrap sample S_3 of 5 data instances to construct Decision tree 3.

Step 5 (b): Compute the best split feature to be placed at each node. Here, we choose Communication Skills and Practical Knowledge.

Step 5 (c): Construct decision tree using CART shown in Figure 12.4.

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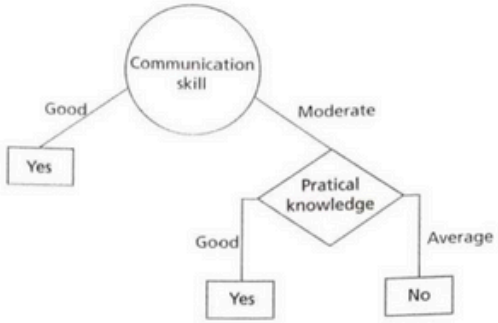

Figure 12.4: Decision Tree 3 with Sample S_3

Step 6: Combine predictions from all classifiers taking average $H_{ave} = \frac{1}{N}\sum_{i=1}^{M} H_i$ for Regression problem or majority voting in the case of classification problem.

Use the data instance 4 as the test instance is not used in the training. The data instance 4 is the Out of Bag instance used for testing.

Following are the predictions given by each of the decision tree constructed using the different samples:

Decision Tree 1 – Yes

Decision Tree 2 – No

Decision Tree 3 – No

It is a classification problem. So, using majority voting, the prediction Job Offer = No is the final prediction of this Random Forest model.

Advantages of the Random Forest Model

1. Can solve both classification problems and regression problems
2. Reduces overfitting and variance compared to a single decision tree
3. Provides more accuracy
4. Performs well even with smaller set of data

Disadvantages of the Random Forest Model

1. Increased complexity in construction and prediction
2. Uses more computational resources

INCREMENTAL ENSEMBLE MODELS

Incremental ensemble model use the predictions that are learnt from the previous leamers features to train a meta learner and make final predictions. Stacking and cascading models an popular incremental ensemble models which are discussed below.

STACKING

Stacking, introduced by Wolpert in 1992, is a meta-modeling technique involving two types of learners: base-learners and a meta-learner. It ensembles multiple heterogeneous models or base learners to make predictions, using these predictions as features to train a second-level meta-learner, whose prediction serves as the final output. Base-learners can be algorithms like Random Forests, SVM, or Perceptron, while the meta-learner, either a meta-regressor or meta-classifier, combines base-learner predictions to enhance model performance by correcting biases. Stacking can also extend to more than two levels, known as multi-level stacking.

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CASCADING

A cascading ensemble model is a multistage learning technique with a sequence of classifiers ordered in terms of increasing complexity and specificity such that early classifiers are simple and general whereas later ones are more complex and specific. It moves to the next stage of learning only if the preceding learner's prediction is not confident. In cascaded models, different classifiers are coupled with their input/output function in a cascade, improving performance at each level. In a typical cascading ensemble model, the complexity of the model increases as more models are added to the cascade